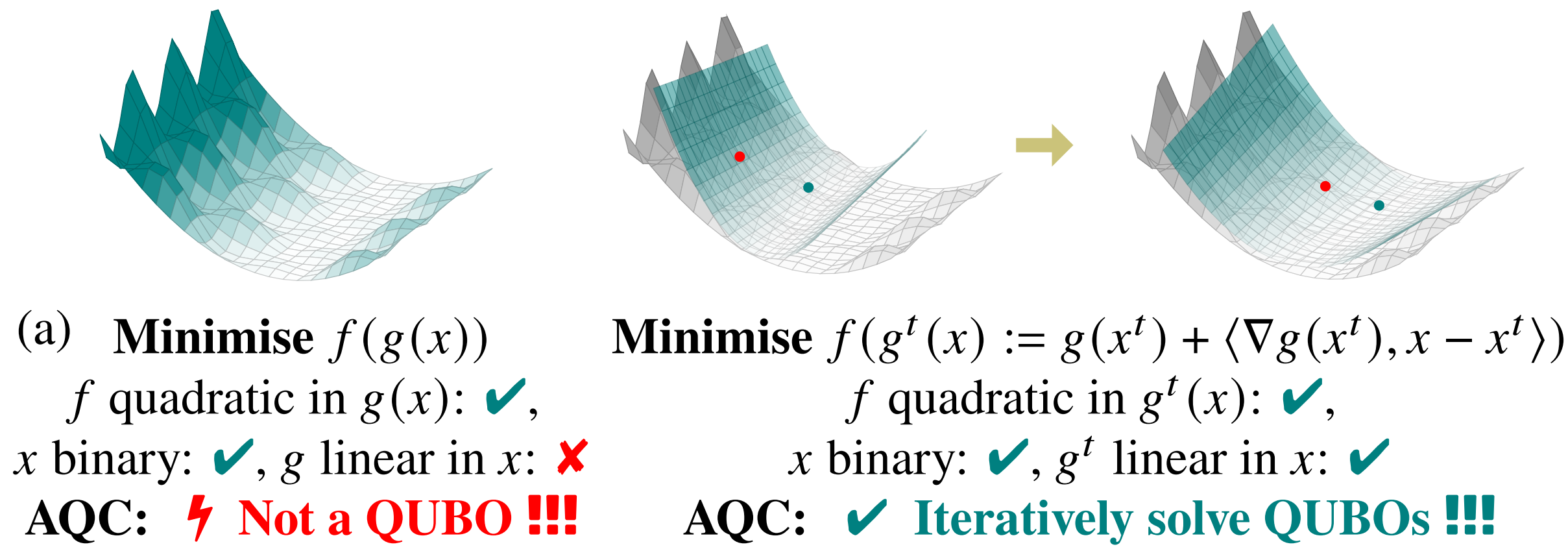


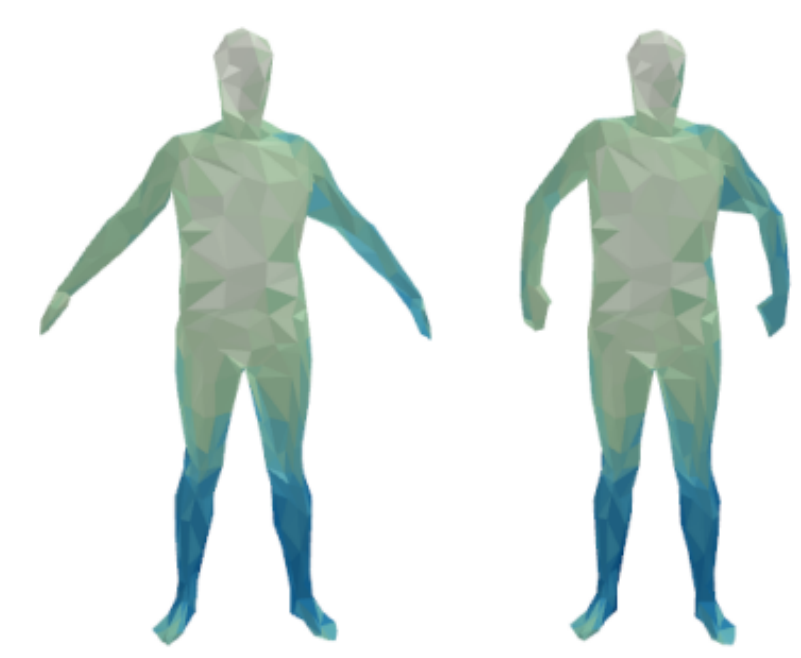
QuCOOP: A Versatile Framework for Solving Composite and Binary-Parametrised Problems on Quantum Annealers

Motivation

Solve $\arg \min_{x \in \{0,1\}^k, g(x) \in \mathcal{S}} f(g(x)), \quad (1)$ **Challenge:** Problem (1) is discrete.
Idea: It may be quantum compatible.
 $f(g(x)) = \langle g(x), \mathbf{Q}g(x) + \mathbf{c} \rangle.$ **Challenge:** It may not be a QUBO.



Example Applications



Shape matching:

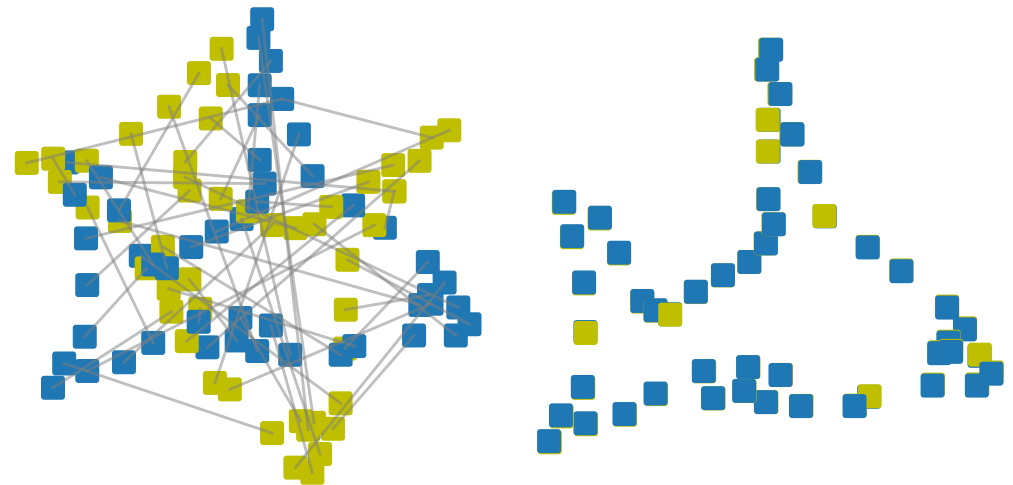
$$\arg \min_{\mathbf{P} \in \Pi_n} \|\mathbf{AP} - \mathbf{PB}\|_F^2, \quad (\text{SM})$$

where $\mathbf{P}$ is a function of binary variables.

Point set registration:

$$\arg \min_{\mathbf{R} \in \text{SO}(d), \mathbf{P} \in \Pi_n} \|\mathbf{XP} - \mathbf{RY}\|_F^2, \quad (\text{PSR})$$

where $\mathbf{R}, \mathbf{P}$ are functions of binary variables.



Binary Parametrisation and QUBO Mapping

Binary parametrisation of permutations: Any permutation matrix $\mathbf{P} \in \Pi_n$ can be parametrised with a length- k , $k = n(n-1)/2$, binary vector $x := (x_1, \dots, x_k)$ via the function

$$\mathbf{P}(x) := \prod_{i=1}^k \mathbf{P}_i(x_i), \quad \mathbf{P}_i(x_i) := \mathbf{T}_i^{x_i},$$

where $\mathbf{T}_i$ is the i th transposition from $((a, b), a, b = 1, \dots, n, a < b)$.

Illustration: $(c, d, b, a) \rightarrow (a, d, b, c) \rightarrow (a, b, d, c) \rightarrow (a, b, c, d)$

Fix position 1 Transpose (1, 4) Fix position 2 Transpose (2, 3) Fix position 3 Transpose (3, 4) Done

$$\mathbf{P} = (3, 4)^1 \cdot (2, 4)^0 \cdot (2, 3)^1 \cdot (1, 4)^1 \cdot (1, 3)^0 \cdot (1, 2)^0$$

Binary parametrisation of rotations: Any rotation matrix in $\text{SO}(d)$ can be parametrised as

$$\mathbf{R}(y) = \exp(\mathbf{M}(y)), \quad \mathbf{M}(y) = \begin{pmatrix} 0 & -y_3 & y_2 \\ y_3 & 0 & -y_1 \\ -y_2 & y_1 & 0 \end{pmatrix} \text{ in } \mathbf{2,3D}.$$

Trick: Write $y_i = a \cdot \sum_j 2^j q_j$, $q_j \in \{0, 1\}$ and refine interval:

$$\begin{matrix} [++++] \\ y_i^{t+1} & y_i^t \end{matrix} \rightarrow \begin{matrix} [++++] \\ y_i^t \end{matrix}$$

QuCOOP as Applied to SM and PSR

Set $\mathbf{p}^t(x) = \mathbf{p}(x^t) + \langle \nabla \mathbf{p}(x^t), x - x^t \rangle$ and $\mathbf{r}^t(y) = \mathbf{r}(y^t) + \langle \nabla \mathbf{r}(y^t), y - y^t \rangle$.

In each iteration t :

(SM):

$$\arg \min_{x \in \{0,1\}^k} \langle g^t(x), \mathbf{Q}g^t(x) \rangle + \square,$$

$$\mathbf{Q} := \alpha \mathbf{I}_{n \times n} - \mathbf{A} \otimes \mathbf{B},$$

With

$$\begin{aligned} g^t(x) &:= \mathbf{p}^t(x) && \text{in (SM)} \\ g^t(x, y) &:= (\mathbf{p}^t(x), \mathbf{r}^t(y)) && \text{in (PSR)} \\ \square &:= \alpha \cdot \|(g^t(\cdot))\|^2, \quad \alpha \in \mathbb{R}_+. \end{aligned}$$

(PSR):

$$\arg \min_{(x,y) \in \{0,1\}^k} \langle g^t(x, y), \mathbf{Q}g^t(x, y) \rangle + \square,$$

$$\mathbf{Q} := \begin{pmatrix} \alpha \mathbf{I}_{n \times n} & -\frac{1}{2} \mathbf{X} \otimes \mathbf{Y} \\ -\frac{1}{2} (\mathbf{X} \otimes \mathbf{Y})^\top & \beta \mathbf{I}_{d \times d} \end{pmatrix},$$

QuCOOP: COmposite OPTimisation on QA

Initialize x^0 and repeat for $t = 0, 1, 2, \dots$

$$g^t(x) := g(x^t) + \langle \nabla g(x^t), x - x^t \rangle \quad (2)$$

$$x^{t+1} \leftarrow \arg \min_{x \in \mathbb{B}^k} \{f^t(x) := f(g^t(x))\} \quad \text{s.t.} \quad g^t(x) \in \mathcal{S} \quad (3)$$

Equivalent QUBO

In each iteration t , solve

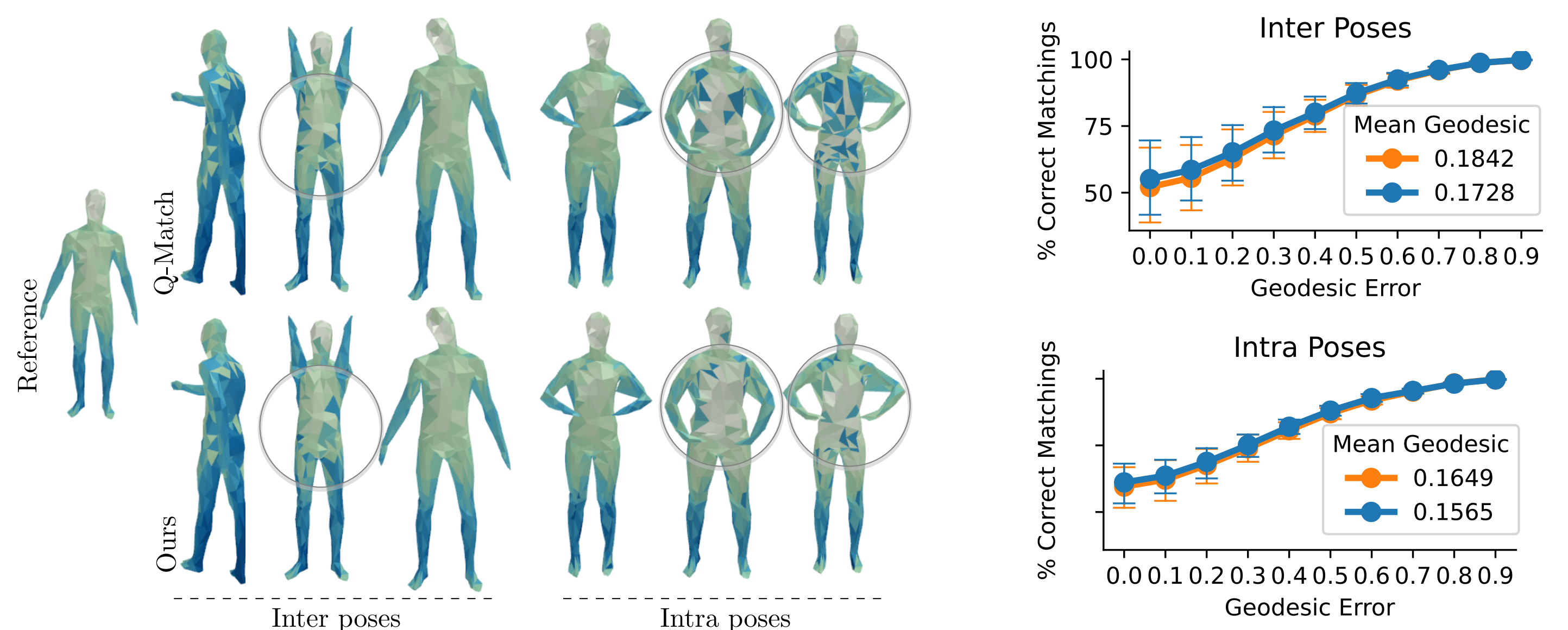
$$\arg \min_{x \in \{0,1\}^k} \langle x, \mathbf{Q}^t x + \mathbf{c}^t \rangle + \text{PenaltyTerm}(g(x^t)),$$

$$\mathbf{Q}^t := \nabla g(x^t) \mathbf{Q} \nabla g(x^t)^\top,$$

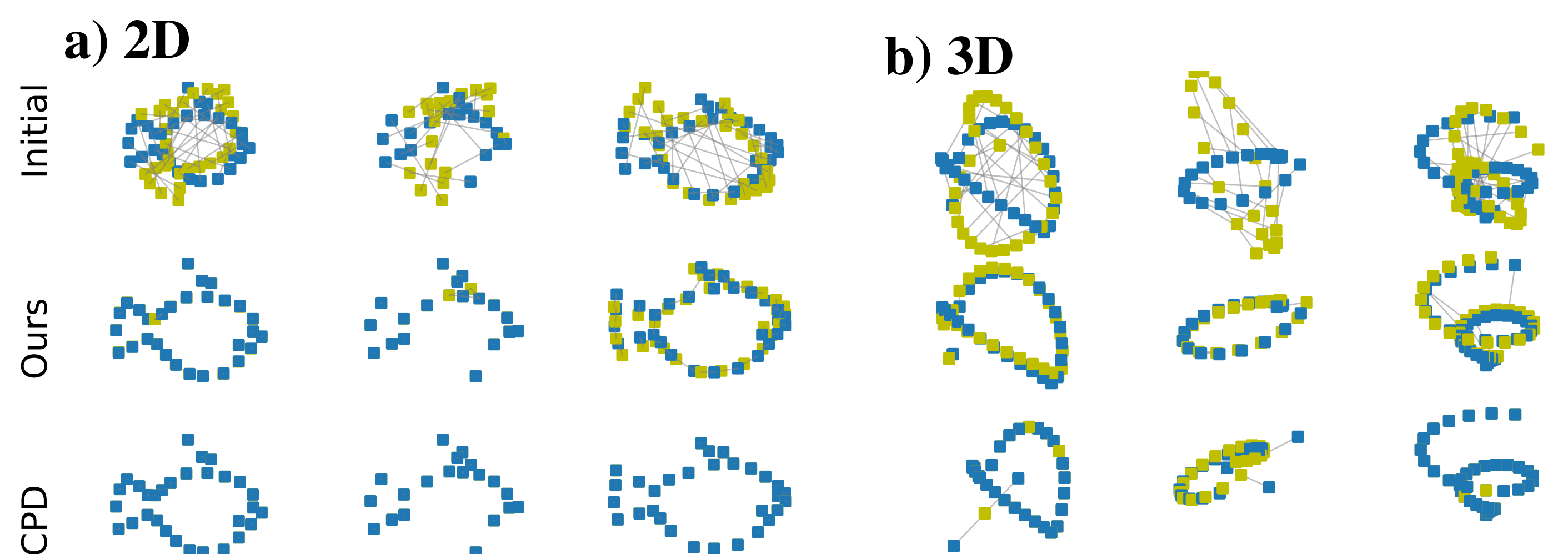
$$\mathbf{c}^t := \nabla g(x^t) \left(\mathbf{c} + 2\mathbf{Q} \left(g(x^t) - \nabla g(x^t)^\top x^t \right) \right).$$

Results

I. Shape Matching using Simulated Annealing (vs. Q-Match [1])

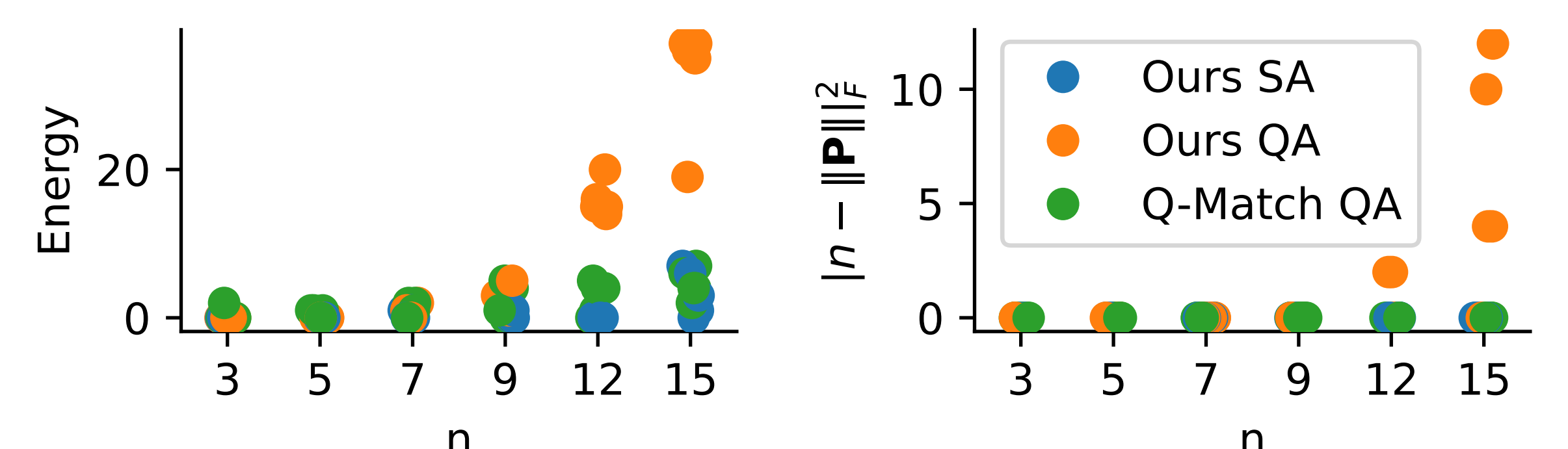


II. Point Set Registration using Simulated Annealing (vs. CPD [2])

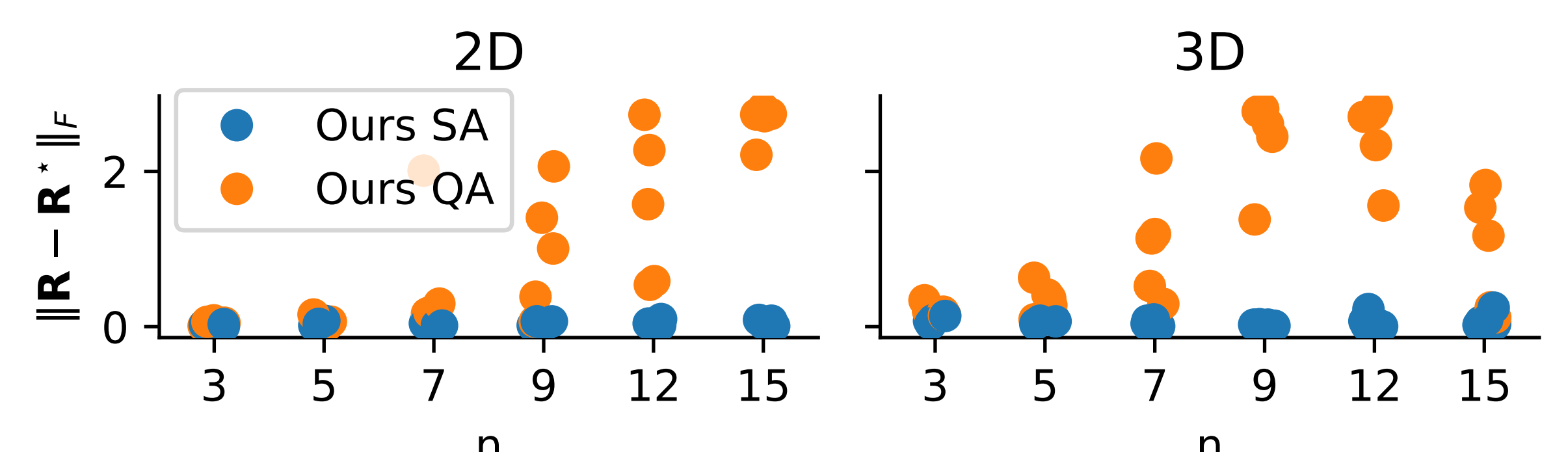


III. Results on D-Wave Quantum Annealers [3]

a) SM



b) PSR



References

- [1] Benkner, Marcel Seelbach and Löhner, Zorah and Golyanik, Vladislav and Wunderlich, Christof and Theobalt, Christian and Moeller, Michael. Q-match: Iterative shape matching via quantum annealing. *International Conference on Computer Vision*, 2021.
- [2] Myronenko, Andriy and Song, Xubo. Point Set Registration: Coherent Point Drift. *PAMI*, 2010.
- [3] D-Wave Systems. QPU Solver Datasheet. https://docs.dwavesys.com/docs/latest/doc_qpu.html, 2024.